CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 5: Database Triggers for Audit Logging & Integrity Rules

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #5 | Academic Year: 2026-27

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1. Overview & Objective:

Develop BEFORE and AFTER row-level triggers recording changes to sensitive financial records.

2. Practical Tasks & Verification Steps:

1. Create audit table student_grade_audit tracking changed_by, changed_at, old_grade, and new_grade.
2. Construct AFTER UPDATE row-level trigger on student_grades table firing for each row change.
3. Construct BEFORE INSERT trigger preventing enrollment if course capacity limit is reached.
4. Execute updates on student records and verify automated immutable rows written to audit log table.

Submission Guidelines & Verification:

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